Finesse x Citadel — Round 2 Portfolio Construction Challenge

Team Name:	MVM Bois
Team Members:	T Abishek, Jason Daniel, Sachith J

1. Objective and Strategy Overview

This report presents a fully systematic, rule-based equity portfolio strategy for the eligible universe of Nifty 100, Nifty Midcap 100, and Nifty Smallcap 100 constituents. The strategy makes no discretionary or judgmental stock selections; every decision — which stocks to hold and how much capital to allocate to each — is produced mechanically from historical price data, so that the process can be reproduced on any date, including data outside the original backtest window.

Every quarter, the strategy computes a composite score for each eligible stock based on two well-documented equity factors: 6-month price momentum (skipping the most recent month to avoid short-term reversal effects) and realised volatility over the same window. Stocks are ranked by this composite score, the top 10 are selected, and capital is allocated inversely to each stock's volatility — steadier stocks receive a larger allocation than more volatile ones, adding a simple, transparent layer of risk control beyond stock selection alone.

2. Data and Investment Universe

Daily adjusted closing prices (split- and dividend-adjusted) were sourced via the Yahoo Finance API (yfinance) for the period 1 January 2021 to 30 June 2026. The eligible universe was constructed from the current constituent lists of Nifty 100, Nifty Midcap 100, and Nifty Smallcap 100, published by NSE Indices.

- 299 of the 300 combined constituent tickers were successfully sourced.
- ZYDUSLIFE.NS could not be retrieved, likely due to its 2022 corporate rename from Cadila Healthcare Ltd, and was excluded from the eligible universe for the full backtest period.
- Current (not historical point-in-time) index constituent lists were used, given project time constraints; this is disclosed as a limitation in Section 8.

3. Methodology

3.1 Signal Construction

For each stock, as of each rebalance date, two quantities are computed over the preceding 6 months: (i) Momentum — the price return from 6 months prior to 1 month prior (i.e. skipping the most recent month), and (ii) Volatility — the standard deviation of daily returns over the full 6-month window. Both are standardised into z-scores across the eligible universe on that date, so that two differently-scaled measures become directly comparable. The composite score is calculated as: Score = Momentum Z-score − Volatility Z-score. A higher score indicates a stock with a stronger recent price trend and comparatively lower volatility.

3.2 Stock Selection

On each rebalance date, all eligible stocks are ranked by composite score in descending order, and the top 10 are selected for the portfolio. No manual override or judgmental adjustment is applied at any stage.

3.3 Portfolio Weighting

Capital is allocated across the 10 selected stocks in inverse proportion to each stock's trailing 6-month volatility: $\text{Weight}_i = (1 / \text{Volatility}_i) / \Sigma(1 / \text{Volatility}_j)$. This ensures that within the selected top 10, calmer stocks receive proportionally larger allocations than more volatile ones, without requiring any subjective judgement.

3.4 Rebalancing and Transaction Costs

The portfolio is rebalanced quarterly (18 rebalance dates across the backtest period). At each rebalance, the strategy computes the rupee value that must be traded for every stock (the absolute difference between its current holding value and its new target value, summed across all stocks entering, exiting, or being resized). A transaction cost of 0.1% is applied to this total traded value at every rebalance, covering both new positions and exits. Over the full backtest, total transaction costs paid were ₹5,44,262 — approximately 2.15% of final portfolio value.

3.5 Economic Rationale

The two factors underlying this strategy were chosen for reasons independent of their performance on this specific dataset, which is a deliberate design choice to reduce the risk of overfitting to 2021-2025 market conditions.

Momentum (with the most recent month skipped) reflects two well-documented, distinct market phenomena operating at different horizons: prices that have trended over 3-12 months tend to continue trending, plausibly due to gradual information diffusion and slow institutional reaction, while prices in the most recent few weeks are more prone to short-term reversal as early movers take profits. Skipping the most recent month is a standard construction (closely related to the academic '12-minus-1' momentum factor) intended to capture the former effect while avoiding the latter.

Volatility was incorporated for two complementary reasons. First, academic literature has documented a persistent 'low-volatility anomaly', in which historically calmer stocks have delivered competitive or superior risk-adjusted returns relative to their more volatile counterparts. Second, and more directly, penalising high volatility in the composite score and using it to size positions reduces the strategy's exposure to the sharp reversals that pure momentum strategies are known to be susceptible to during periods of rapid market regime change.

4. Backtest Results (1 July 2021 – 31 December 2025)

The strategy requires a full 6-month lookback window before its first signal can be computed; consequently, the invested backtest period begins on 1 July 2021 rather than 1 January 2021 (a burn-in period), and runs through 31 December 2025. This is disclosed as a modelling limitation in Section 8.

Metric	Your Strategy	Nifty 100 Benchmark
Starting Capital	₹1,00,00,000	₹1,00,00,000
Final Value	₹2,53,09,580	₹1,66,26,309
Total Net PNL	₹1,53,09,580	₹66,26,309
Total Return	153.35%	66.26%
CAGR (Annualised Return)	23.40%	12.26%
Sharpe Ratio (0% risk-free)	1.22	0.91
Maximum Drawdown	-21.36%	-17.56%

The strategy delivered a CAGR roughly double that of the Nifty 100 benchmark, with a materially higher Sharpe ratio (1.22 vs. 0.91) — indicating better risk-adjusted performance, not merely higher risk-taking. The strategy's maximum drawdown (-21.36%) is somewhat deeper than the benchmark's (-17.56%), a reasonable and expected consequence of holding a concentrated 10-stock portfolio rather than a diversified 100-stock index.

Figure 1: Strategy portfolio value over the backtest period, generated from actual daily simulation output.

5. Trade-Level Statistics

The metrics below describe the strategy's discrete trading behaviour. They are not applicable to the Nifty 100 benchmark, which is held as a single buy-and-hold position with no subsequent trades to measure.

Metric	Your Strategy	Nifty 100 Benchmark
Total Completed Trades	156	N/A (buy-and-hold)
Average Trades per Stock	1.30	N/A
Accuracy (% Profitable Trades)	57.7%	N/A
Average Winning Trade	+21.91%	N/A
Average Losing Trade	-11.24%	N/A
Gain-to-Loss Ratio	1.95	N/A

The strategy's edge is not purely a function of a high hit rate: with an accuracy of 57.7%, slightly better than half of all trades were profitable. The gain-to-loss ratio of 1.95 shows that winning trades were, on average, nearly twice the magnitude of losing trades — meaning the strategy's payoff asymmetry meaningfully contributes to its overall performance, independent of the hit rate alone.

6. Out-of-Sample Stress Test (1 January 2026 – 30 June 2026)

Per the competition guidelines, the same strategy — with no parameter changes or re-tuning of any kind — was extended forward through 30 June 2026 using data outside the original design window. Portfolio value moved from ₹2,53,09,580 (31 December 2025) to ₹2,42,74,674 (29 June 2026), a decline of approximately 4.09% over the six-month stress-test period.

This decline is reported transparently rather than adjusted or hidden. It is consistent with the expected behaviour of a momentum-based strategy during a period of trend reversal or reduced market breadth, and is disclosed here precisely so that the strategy's out-of-sample robustness can be assessed honestly, rather than presenting only the more favourable in-sample figures.

It is worth noting that the strategy's turnover, while resulting in 156 total completed trades, reflects genuine partial persistence rather than a full portfolio reset each quarter — several stocks were carried across consecutive rebalances when their score and volatility profile remained favourable, rather than being mechanically replaced regardless of continued suitability.

7. Portfolio Composition Across All Rebalances

The table below reconstructs the strategy's actual holdings at each of the 18 rebalance dates, derived directly from the submitted code's trade log. At every rebalance, exactly 10 positions were held, consistent with the maximum portfolio size constraint.

Rebalance Date	Stocks Held (10 positions)
2021-07-01	ADANIENSOL, ADANIENT, ATGL, CDSL, COLPAL, GRASIM, PGEL, POLYCAB, POONAWALLA, SUPREMEIND
2021-10-01	ALKEM, ANGELONE, BAJAJFINSV, DIVISLAB, INFY, JSWENERGY, LAURUSLABS, SBILIFE, SRF, WIPRO
2022-01-03	ADANIENSOL, ATGL, BAJAJFINSV, BAJAJHLDNG, CGPOWER, JBMA, JSWENERGY, LODHA, LTM, TECHM
2022-04-01	ADANIGREEN, CGPOWER, INFY, JBMA, KPITTECH, LT, NESTLEIND, NH, PAGEIND, POWERGRID
2022-07-01	ADANIPOWER, BDL, BLS, COLPAL, HAL, IRFC, ITC, M&M, NTPC, SWANCORP
2022-10-03	ABB, ADANIENT, BEL, CASTROLIND, GESHIP, HAL, ITC, M&M, TVSMOTOR, VBL
2023-01-02	ABCAPITAL, ADANIENT, BHEL, COCHINSHIP, GRASIM, ICICIBANK, KARURVYSYA, MAZDOCK, RVNL, UNIONBANK
2023-04-03	BLUESTARCO, CUMMINSIND, HCLTECH, ITC, LT, OIL, ONGC, PERSISTENT, PFC, RECLTD
2023-07-03	ABB, AUROPHARMA, BAJAJ-AUTO, CGPOWER, CHOLAFIN, ITC, KAYNES, OFSS, POLYCAB, TMPV
2023-10-03	BSE, FORCEMOT, HBLENGINE, HSCL, IDFCFIRSTB, LUPIN, MAZDOCK, RECLTD, SUZLON, TRENT
2024-01-01	ANANDRATHI, BSE, HBLENGINE, KALYANKJIL, MCX, PFC, RECLTD, SUZLON, TRENT, WELCORP
2024-04-01	ANANDRATHI, BOSCHLTD, BPCL, CUMMINSIND, GVT&D, INOXWIND, IOC, SIGNATURE, SWANCORP, WOCKPHARMA
2024-07-01	ANANDRATHI, BLUESTARCO, COCHINSHIP, CUMMINSIND, FORCEMOT, GVT&D, HINDZINC, NETWEB, SIEMENS, VEDL
2024-10-01	COLPAL, COROMANDEL, CROMPTON, DIVISLAB, GLENMARK, INFY, LALPATHLAB, PGEL, TRENT, WHIRLPOOL
2025-01-01	COFORGE, COHANCE, DEEPAKFERT, DIVISLAB, HCLTECH, MCX, NEULANDLAB, PAYTM, SARDAEN, SUNPHARMA
2025-04-01	BAJFINANCE, HDFCBANK, ICICIBANK, KOTAKBANK, MUTHOOTFIN, NH, PATANJALI, SBICARD, SHREECEM, SRF
2025-07-01	BAJFINANCE, BDL, BRITANNIA, FORCEMOT, HDFCLIFE, ICICIBANK, MFSL, SBICARD, SBILIFE, SOLARINDS
2025-10-01	ANANDRATHI, BOSCHLTD, DELHIVERY, ETERNAL, FORCEMOT, GVT&D, JMFINANCIL, LTF, MFSL, POLYCAB

A qualitative review of this table shows the portfolio's composition shifting meaningfully over time in response to the momentum-volatility signal, rather than remaining anchored to a fixed sector or theme. Early periods (2021-2022) show a concentration in power, energy, and infrastructure-linked names; mid-period rebalances (2023) show a rotation toward industrials, banking, and PSU-linked names; and later periods (2024-2025) show broader participation from smaller, less liquid names as well as a rotation into private-sector financials (HDFCBANK, ICICIBANK, KOTAKBANK, BAJFINANCE) by early-to-mid 2025. This rotation is a natural consequence of a purely price-driven signal reacting to whichever segment of the market is exhibiting leadership at a given time, and is not the result of any manual sector allocation decision.

8. Assumptions and Limitations

- Current (not historical point-in-time) constituent lists were used for the eligible universe; this may understate survivorship bias relative to a fully point-in-time reconstruction.
- ZYDUSLIFE.NS was excluded from the universe due to a data-sourcing issue, likely related to its 2022 corporate rename.
- The invested backtest period begins 1 July 2021 (not 1 January 2021) due to the 6-month signal lookback requirement.
- Small individual gaps in daily price data were forward-filled rather than treated as missing.
- Fundamental (accounting) data was deliberately not used, in favour of a fully price-based, look-ahead-bias-free signal that can be computed identically on any date.

- Transaction costs of 0.1% were applied to the net rupee value traded per stock per rebalance, consistent with the competition's stated cost assumption.

9. Tools and Software

Python 3.14, with pandas for data handling and yfinance for historical price retrieval. All computations (signal generation, portfolio construction, backtesting, and metrics) are implemented from first principles rather than third-party backtesting libraries, so that every calculation is fully transparent and auditable from the submitted source code.

10. Potential Improvements and Future Work

Given additional time, the following extensions would be prioritised, in order of expected impact on strategy robustness:

- Point-in-time universe reconstruction: sourcing historical Nifty 100/Midcap 100/Smallcap 100 constituent lists as of each rebalance date, rather than using current constituent lists throughout, would eliminate the survivorship bias disclosed in Section 8 and likely produce a more conservative, more realistic backtest result.
- Extended lookback buffer: sourcing price history from mid-2020 rather than January 2021 would remove the 6-month burn-in period, allowing the invested backtest to begin on 1 January 2021 as specified, rather than 1 July 2021.
- A long-term trend filter: incorporating a simple filter such as a 200-day moving average (only holding a stock if its price is above its own long-term trend) is a well-established, easily-defensible risk-management addition that could reduce drawdown during broad market downturns, at the cost of some upside participation.
- A modest quality overlay: supplementing the current price-only signal with a lightweight, point-in-time-safe fundamental filter (e.g. excluding stocks with deteriorating profitability) could reduce the strategy's exposure to momentum 'traps' — stocks that have risen for reasons unrelated to underlying business fundamentals.
- Sensitivity analysis: systematically testing the strategy's performance across a range of lookback windows (e.g. 3, 6, 9, 12 months) and rebalance frequencies (monthly, quarterly, semi-annually), evaluated only on 2021-2025 data, would provide stronger evidence that the chosen parameters are not a narrow, coincidental best-fit to this specific window.

11. Conclusion

The strategy presented here is intentionally simple and fully mechanical: two well-documented price-based factors, combined with a straightforward volatility-based weighting scheme, rebalanced quarterly. It outperformed the Nifty 100 benchmark on both an absolute and risk-adjusted basis over the backtest period, with a transparent, disclosed decline during the out-of-sample stress test. We believe this transparency — including reporting a result that is not universally favourable — reflects a methodology built for robustness and reproducibility rather than one fitted to a specific historical window.